

Effect of covert nutritive dilution on the spontaneous food intake of obese individuals: a pilot study.

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The present study investigated the feasibility of a new experimental approach for studying the effect of covert nutritive dilution on the spontaneous food intake of obese individuals. Eight obese subjects were studied as inpatients on a metabolic unit for 15 days during which time they were unaware that their food intake was being monitored. A platter method of food presentation encouraged ad libitum ingestion. Caloric dilution was achieved by replacing sucrose-containing products with aspartame-sweetened analogues in an otherwise normal diet. During the base-line period the subjects spontaneously ate sufficient conventional food to maintain or even slightly increase body weight. Covert substitution of aspartame-sweetened products for their sucrose counterparts resulted in an immediate reduction in spontaneous energy intake of approximately 25%. The aspartame analogues were as well accepted as their conventional counterparts, as indicated by the equal quantity of each consumed. These preliminary results demonstrate that, in a metabolic ward setting, it is possible to maintain the spontaneous food intake of obese individuals at levels sufficient to preserve body weight and arbitrarily to decrease those levels of intake by 25% or more through covert changes in the caloric density of the diet.